TAXI FARE PREDICTION

Using Random Forest Regressor

1. Project Overview

The objective of this project is to predict taxi fares using trip location, distance, passenger count, and pickup-time information. A Random Forest Regressor was trained, evaluated, and tuned to improve prediction performance.

2. Dataset Description

The TaxiFare.csv dataset contains 50,000 records and 8 original columns, including pickup and drop-off coordinates, pickup date and time, passenger count, and fare amount. The target variable is amount.

3. Data Preprocessing

- Checked missing values and duplicate records; none were found in the original dataset.
- Removed invalid coordinates, non-positive fares, zero-passenger records, and selected suspicious geographic or fare records.
- Excluded unique_id because it is an identifier rather than a useful predictive feature.
- Retained plausible zero-distance trips; zero distance alone was not treated as proof of an invalid trip.
- Removed the diagnostic fare_per_km column before modeling because it is undefined for zero-distance trips.

After cleaning, the dataset contained 48,731 rows. The geographic filtering used approximate New York City boundaries, so valid trips outside those boundaries may also have been excluded.

4. Exploratory Data Analysis

- Average trip distance: 3.33 km; median: 2.16 km; maximum: 43.65 km.
- Average fare: approximately $11.3. The maximum retained fare was $104.67.
- Trip distance had a strong positive correlation with fare (approximately 0.732).
- Passenger count had a very weak correlation with fare (approximately 0.016).
- Average weekday and weekend fares were similar: approximately $11.34 and $11.29.

5. Feature Engineering

The Haversine formula was used to calculate trip distance from pickup and drop-off coordinates. The following time-based features were created:

- pickup_hour, pickup_day, pickup_month, and pickup_weekday
- is_weekend — identifies Saturday and Sunday
- is_night — identifies pickup times from 10:00 PM to 4:59 AM

6. Model Development

The target variable was amount. Twelve input features were used: pickup and drop-off coordinates, passenger count, trip distance, pickup hour/day/month/weekday, is_weekend, and is_night. The data was split into 80% training and 20% testing using random_state=42.

A Pipeline containing StandardScaler and RandomForestRegressor was used. Scaling is not required by Random Forest, but placing it inside the pipeline keeps preprocessing fitted only on the training folds and satisfies the project workflow.

7. Baseline Model Results

Metric	Baseline Result
MAE	2.1489
MSE	16.3723
RMSE	4.0463
R^2 Score	0.8196

8. Hyperparameter Tuning

GridSearchCV was used with 3-fold cross-validation and negative mean absolute error as the scoring metric. The search tested n_estimators (100, 200), max_depth (10, 15), and min_samples_split (2, 5).

Best parameters: n_estimators=200, max_depth=15, and min_samples_split=5. The best cross-validation MAE was approximately 2.1106.

9. Tuned Model Evaluation

Metric	Tuned Result
MAE	2.1354
MSE	16.2539
RMSE	4.0316
R^2 Score	0.8209

The tuned model performed slightly better than the baseline model across the reported test metrics. Its five-fold cross-validation MAE was 2.0984 (standard deviation 0.0339), indicating relatively consistent validation performance.

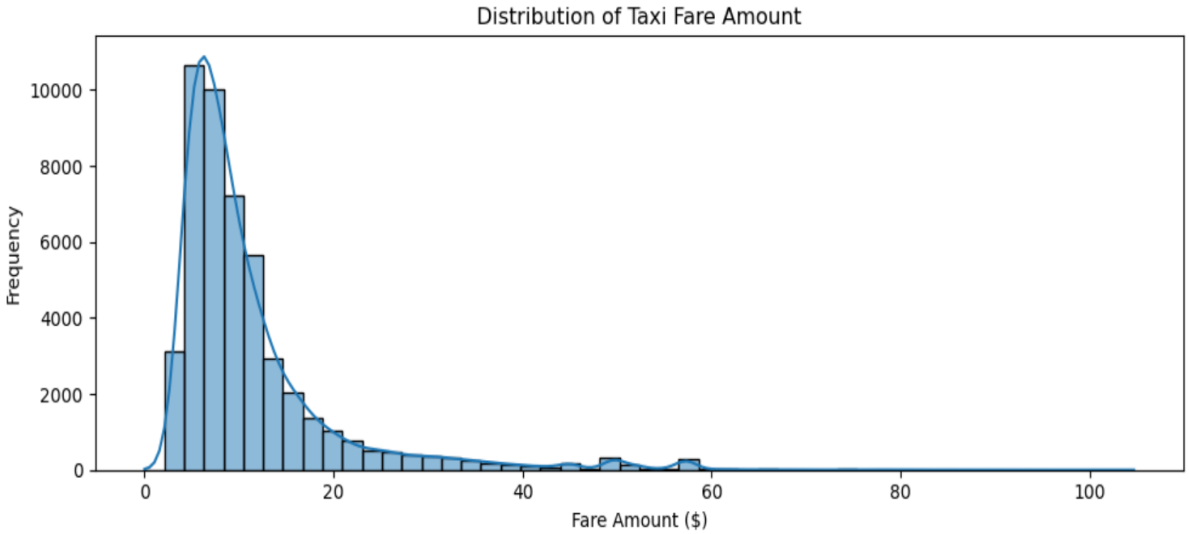
10. Feature Importance

Trip distance was the dominant feature, contributing approximately 85.41% of the model's impurity-based feature importance. Drop-off and pickup longitude/latitude were the next most influential features. Feature importance indicates model reliance, not causation, and correlated inputs can affect these values.

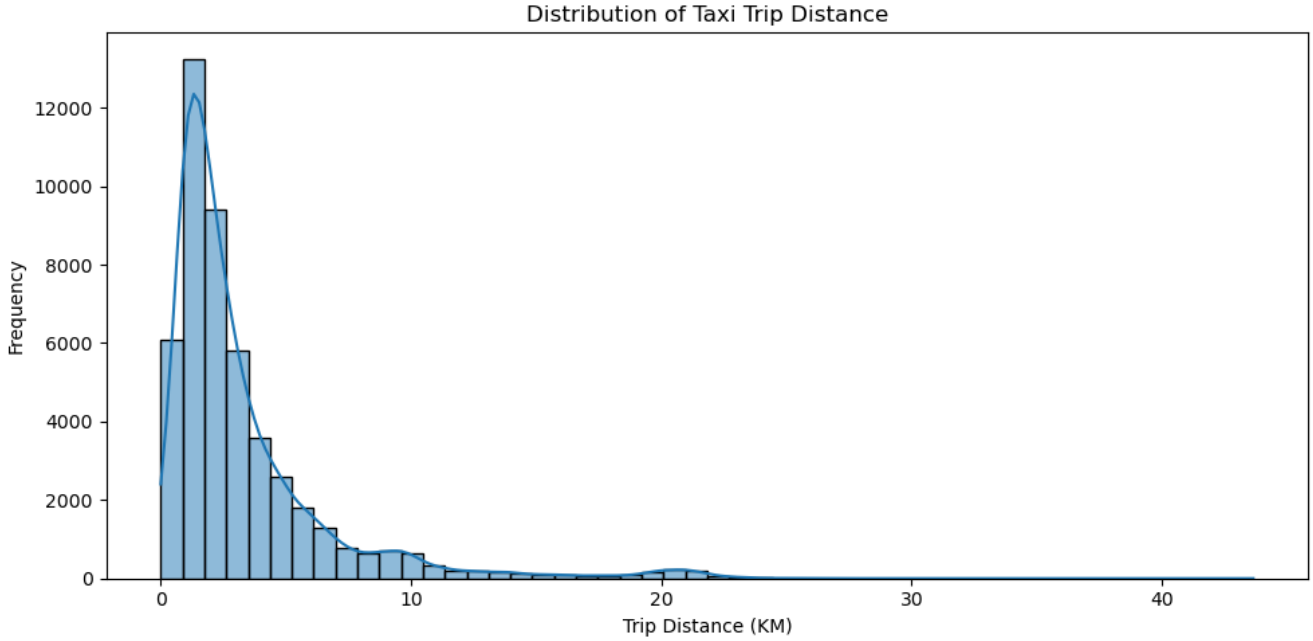
11. Model and Prediction Export

The trained model was saved as taxi_fare_random_forest.pkl using joblib. Predictions were exported to taxi_fare_predictions.csv with Actual_Fare, Predicted_Fare, and Absolute_Error columns.

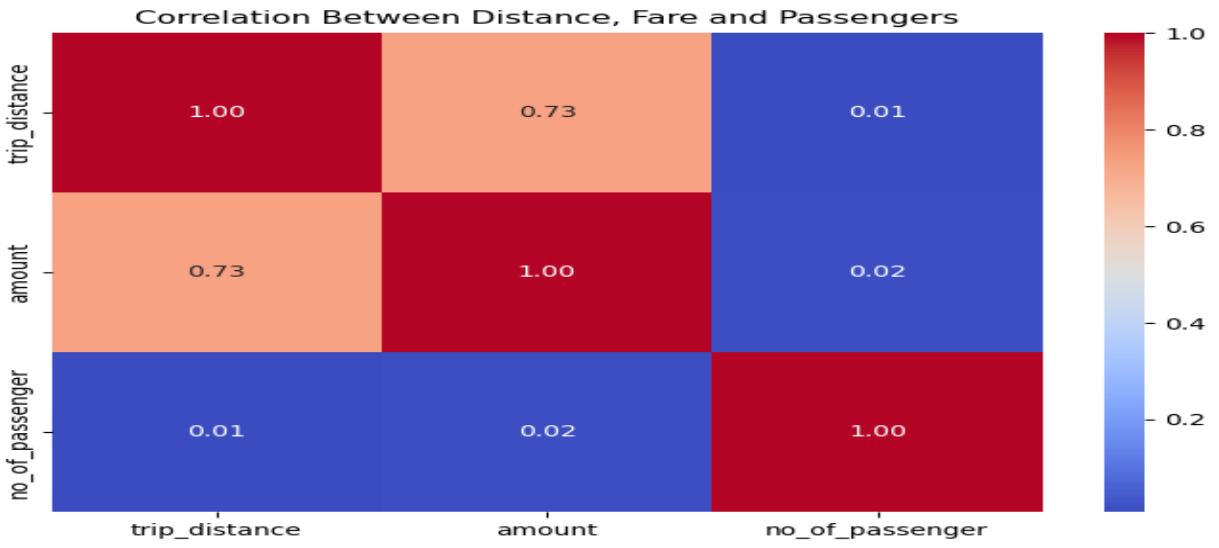
1. Taxi fare distribution



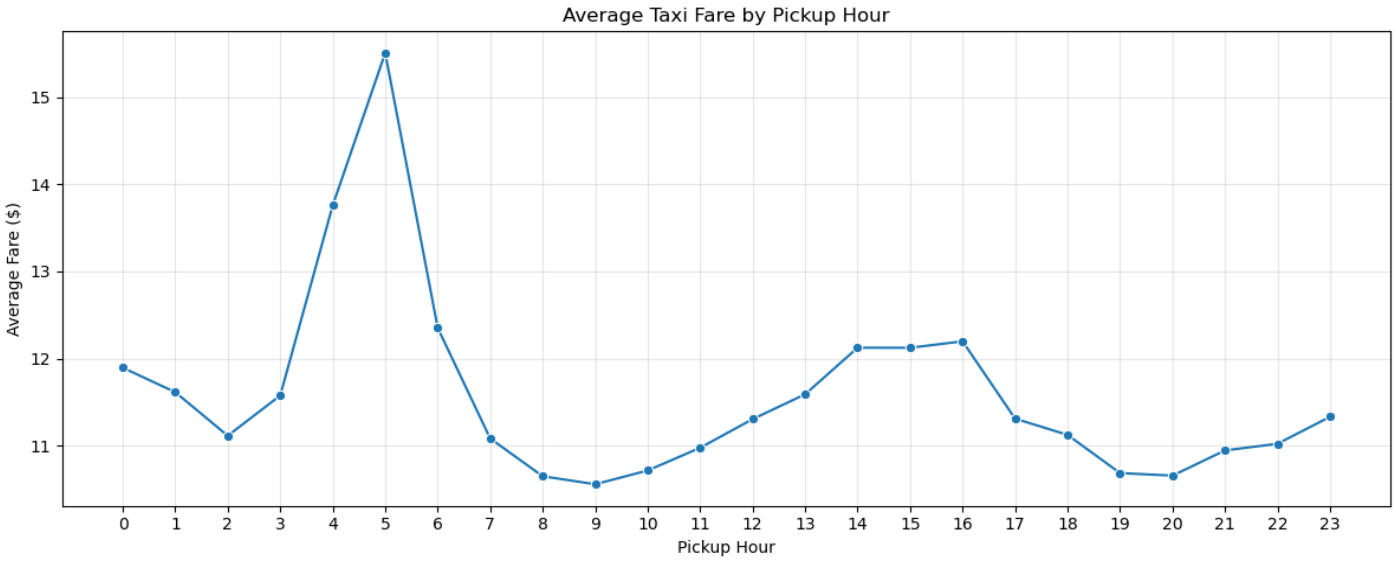
2. Trip distance distribution



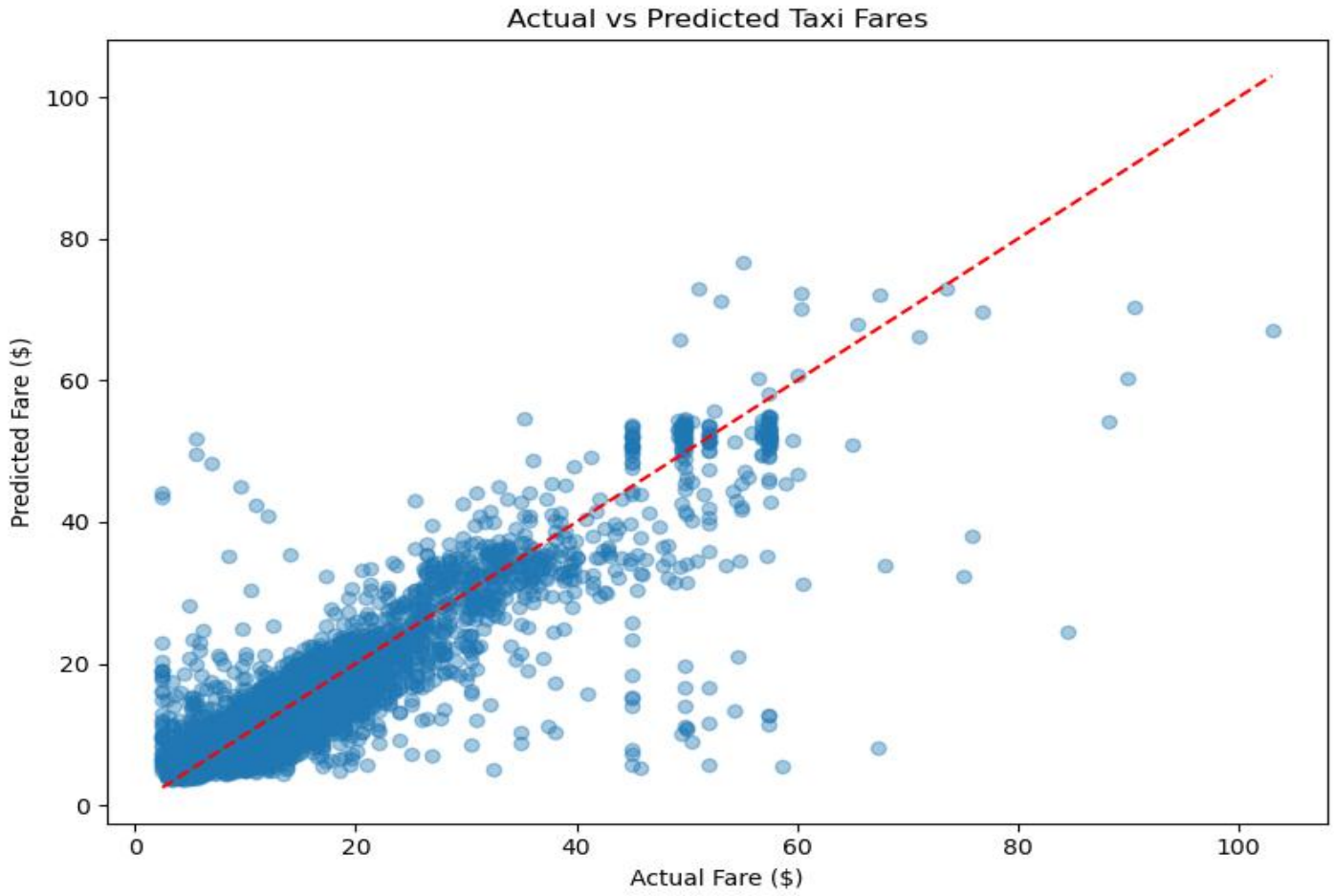
3. Correlation heatmap



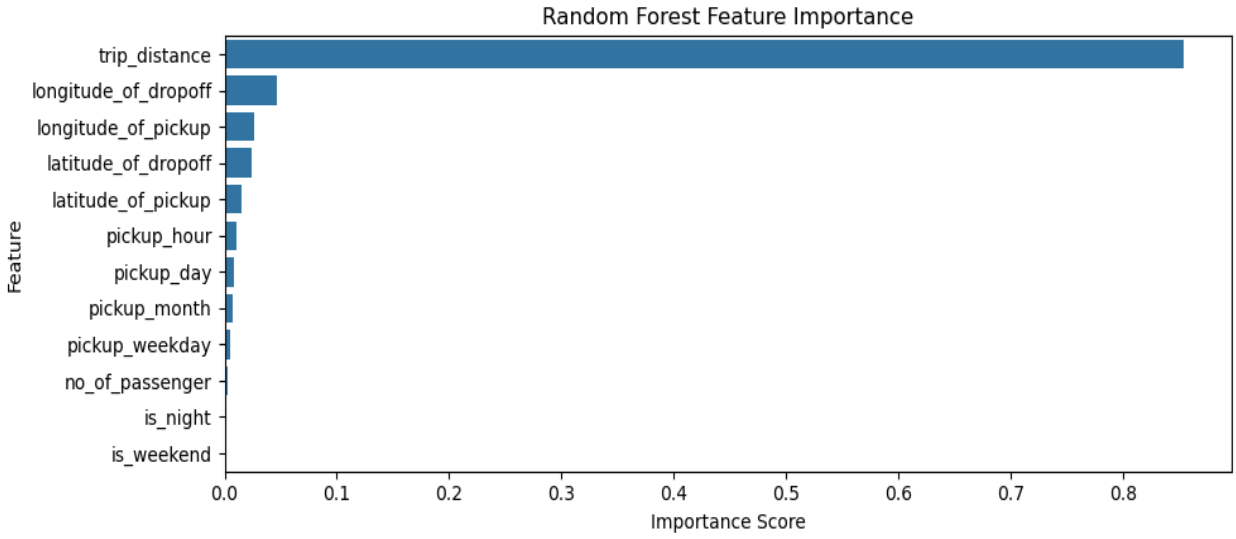
4. Average fare by pickup hour



5. Actual vs predicted fares



6. Random Forest feature importance



12. Limitations

- The dataset does not contain a drop-off timestamp, so actual trip duration could not be calculated.
- Approximate geographic boundaries may exclude legitimate trips outside the selected New York City area.
- The model is trained on historical data and may not reflect current fare rules, surcharges, or traffic conditions.
- Feature importance should not be interpreted as a causal explanation.

13. Conclusion

The Random Forest Regressor achieved a test R^2 score of 0.8209 and an MAE of 2.1354 after hyperparameter tuning. Trip distance was the most influential predictor of taxi fare. The project demonstrates a complete regression workflow covering data cleaning, feature engineering, model training, hyperparameter tuning, evaluation, and export of predictions.